Introduction to Programming

Fundamentals of Programming – CSC10012

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Contents

- Computer programming.
- Computer algorithm.

Programming Concepts

- Programming concepts:
  - Problem:

    Teach a kid to find sum of max and min among 7, 1, 9.

    Rules:

    The kid knows addition and comparison between two numbers.

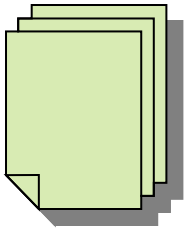
    Write solution in step-by-step.
  - Programming ~ teaching:

    From what computer knows.

    Step-by-step.

Computer

Programming



Program

Programming Concepts

- Programming concepts:
  - Computer has a set of hard-wired instructions.
  - Programming: write instructions for computer.
  - Computer program:

    Written organized instructions.

    Follow step-by-step.

    To solve problem.

Software Engineer

- Programmer:
  - Person who writes the program.
  - Programmer, coder, developer.
  - To become a good programmer:
    - Logical thinking.
    - Analytical and careful.
    - Hands-on and explore.
  - The first programmer?

Ada Lovelace

Instruction Code

- Instruction code:
  - Machine code:
    Binary sequence of ‘0’ and ‘1’.
    Computer can execute directly.
  - Pseudo-code:
    Natural language.
    Human can understand.
  - Source code:
    Intermediate language.
    Easy to read, not too vague.
    Can be translated to machine code.
      Compiler vs. Interpreter.
      Who wrote the first compiler?

Step 1: add a, b.
Step 2: compare a, c.
Step 3: output a.
…

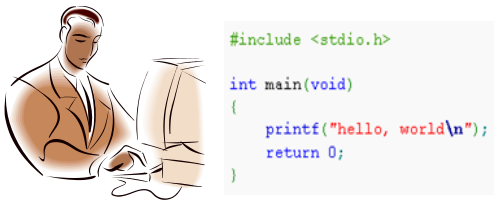
```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
    return 0;
}
```

Programming Cycle

- Programming cycle:
  - Edit: write source code.
  - Compile: source code → machine code.
  - Run: execute machine code to perform tasks.
  - Debug: review source code to fix errors.

➔ Debug takes the most time!



```
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
    return 0;
}
```

Edit / Debug

Compile

Run

Programming Language

- Programming language:
  - Intermediate language between human and computer.
  - Concise and precise.
  - There are thousands of programming languages!
  - Learning programming thinking is more important than learning programming language.

Programming Language

- Programming language:
  - Low-level language:

    1st generation: machine code.

    2nd generation: assembly language.
  - High-level language:

    Procedural: Fortran, Pascal, C.

    Functional: Lisp, Haskell, F#.

    Object oriented: C++, Java, C#.
  - Trends:

    Scripting languages: JavaScript, Python.

    AI prompts: ask AI to generate source code.

Programming Environment

- Programming environment:
  - A platform of tools supporting programming tasks.
  - Source code editor:

    Edit text, syntax highlight, code completion.
  - Compiler:

    Translate source code to machine code.
  - Terminal:

    Command-line tool to run program and shell commands.
  - Debugger:

    Run program step-by-step.

    Monitor computer memory.

Programming Environment

- Programming environment:
  - IDE – Integrated Development Environment:

    A single application combining programming tools all-in-one.

    User friendly, configuration-free, add-in features.

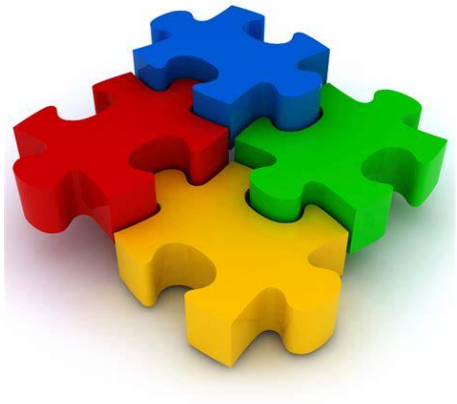
    Cons:

    Heavy memory and storage usage.

    Limit configuration skill.

    Require license.

    Track user data.



Programming Environment

- C/C++ programming environment:

Tools	Windows	MacOS	Linux
Editor	Notepad++ VS Code (*) Sublime Text (*)	TextMate VS Code (*) Sublime Text (*)	Vim, Emacs VS Code (*) Sublime Text (*)
Compiler	MSVC (*) GCC (+) Clang (+)	Clang GCC	GCC Clang
Debugger	MSVC (*) GDB	LLDB GDB	GDB LLDB
IDE	Visual Studio (*) Code::Blocks CodeLite	Xcode (*) Code::Blocks CodeLite	Code::Blocks CodeLite

(*): Proprietary software and track user data.
(+): Run on Linux simulated environment like MinGW64, WSL.

Programming Environment

- C/C++ programming environment:
  - Set up loosely-coupled environment:

    Windows: MinGW64, GCC, GDB.

    Project WinLibs: light-weight, not extensible, https://winlibs.com

    Project MSYS2: get MSYS2 and toolchain, https://www.msys2.org

    Linux (Ubuntu/Debian): GCC, GDB.

    Install package **build-essential**.

    MacOS: Clang, LLDB.

    Terminal command: **xcode-select --install**.
  - Set up IDE:

    Visual Studio (Windows): https://visualstudio.microsoft.com

    Xcode (MacOS): xcode-select --install, then Get Xcode.

    Code::Blocks: https://www.codeblocks.org

    CodeLite: https://codelite.org

Contents

- Computer programming.
- Computer algorithm.

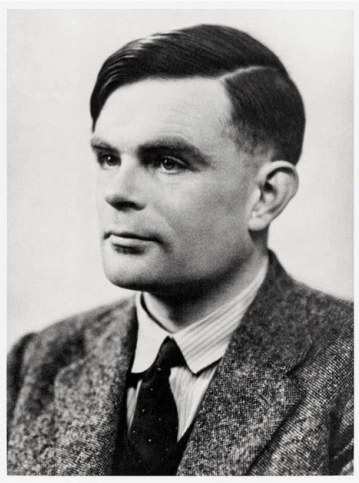
Algorithm Concepts

- Algorithm concepts:
  - Different machines have different instruction code.
  - Turing machine: abstract machine and instructions.
  - Algorithm ~ program on Turing machine:

    Finite steps to solve problem on Turing machine.

    Does not depend on specific machine.

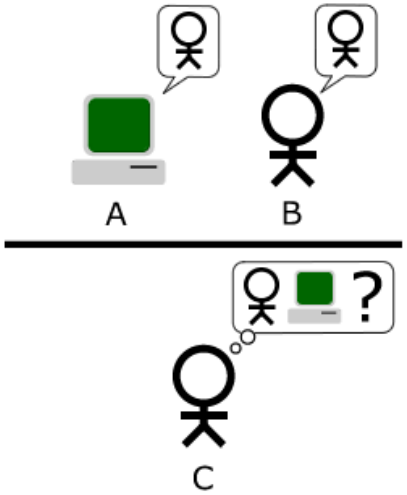
    Is more abstract than program.



Alan Turing

Basic Instructions

- Algorithm basic instructions:
  - Read input, write output.
  - Arithmetic operations: +, -, *, /, =.
  - Comparisons: >, <, >=, <=, ==, !=.
  - Condition: if-else.
  - Loop: while.
- Can machines think?
  - The Imitation game - Turing test.



Notations

- Algorithm representations:
  - Pseudo-code:

    Use natural language.

Instructions	Notations	Example
Read input Write output	Input Output	-Step 1: **Input** a, b, c. -Step 2: **Output** a + b + c.
Arithmetic operations	+, -, *, /, %, =	-Step 1: a = 5. -Step 2: b = (a + 5) / (a – 3).
Condition	If <condition> … Else …	-Step 1: **Input** a. -Step 2: **If** a < 0 **Output** “Negative” **Else Output** “Positive”.
Loop	While <condition> …	-Step 1: a = 0. -Step 2: **While** a < 10 **Output** a. a = a + 1.

Notations

- Algorithm representations:
  - Pseudo-code:

* Find max among 3 numbers:

-Step 1: **Input** a, b, c.

-Step 2: max = a.

-Step 3: **If** b > max
max = b.

-Step 4: **If** c > max
max = c.

-Step 5: **Output** max.

* Compute factorial of N:

-Step 1: **Input** N.

-Step 2: S = 1.

-Step 3: **While** N > 1
S = S * N.
N = N – 1.

-Step 4: **Output** S.

Notations

- Algorithm representations:
  - Flowchart:

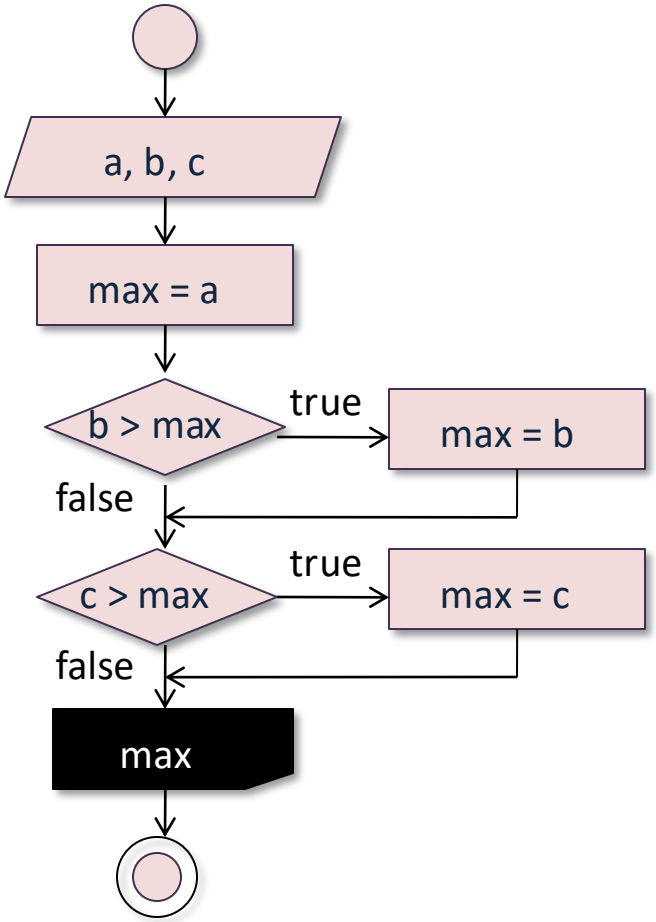
    Use diagram.

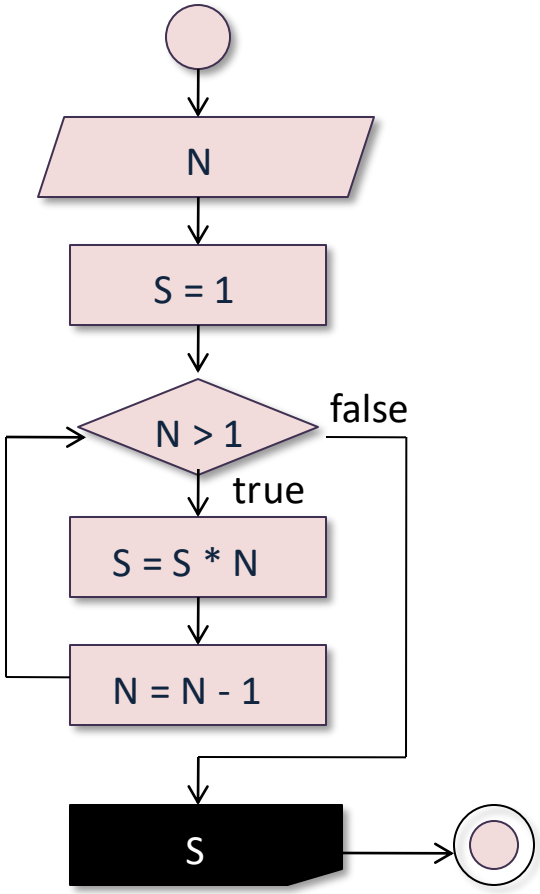
    Visualize flow of process.

Intrustions	Notations	Example
Read input Write output		a, b, c x, y
Arithmetic operations		a = a + b
Condition		a < 0
Transition		

Examples

- Algorithm representations:





Complexity

- Algorithm efficiency:
  - Complexity:

    Algorithm performance varies with different inputs.

    Big-O notation: O(), growth rate.

    Time complexity: input ~ instructions.

    Space complexity: input ~ memory.

    Classes of complexity:

    Constant: O(1).

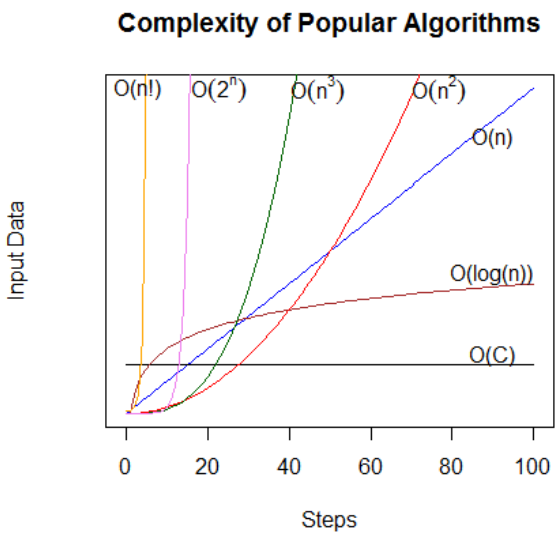
    Logarithmic: O(log(n)).

    Linear: O(n).

    Polynomial: O(n^2), O(n^3), ...

    Exponential: O(2^n).

    Factorial: O(n!).



Complexity

- Algorithm efficiency:
  - Popular algorithm complexities:

Algorithm	Complexity	Form
Finding max (3 numbers)	O(1)	Constant
Binary Search(N elements)	O(log(n))	Logarithmic
Factorial N!	O(n)	Linear
Sorting (N elements)	Bubblesort: O(n^2) Quicksort: O(n*log(n))	Polynomial
Traveling Salesman (N city)	O (K^n) O(n!)	Exponential Factorial

  - Simplicity:

    Easier to understand and implement.

    Reduce cost in debugging and maintenance.

Summary

- Computer programming:
  - Write instructions for computer to solve problem.
  - Programming cycle: edit, compile, run, debug.
  - Programming language:

    Intermediate language to communicate with computer.

    Low-level vs high-level.

    Trends: scripting languages, AI prompts.
  - Programming environment:

    Platform of tools supporting programming tasks.

    Editor, compiler, terminal, debugger.

    IDE: Integrated Development Environment.

Summary

- Computer algorithm:
  - Finite steps to solve a problem on Turing machine.
  - Turing machine: abstract machine and instructions.
  - Basic instructions:

    Input, output.

    Arithmetic operations, comparisons.

    Condition, loop.
  - Representations: pseudo-code, flowchart.
  - Efficiency:

    Complexity: Big-O notation, growth rate.

    Simplicity: easy to understand or implement.

Practice

- Practice 1.1:

  Write algorithm to compute person age as follow:

  - Input person birth-year.
  - Compute and output person age.

Practice

- Practice 1.2:

  Write algorithm to compute lucky number of a car as follow:

  - Input car registration number (a 5-digit number).
  - Compute and output the lucky number.

Practice

- Practice 1.3:

  Write algorithm to simulate a calculator as follow:

  - Input two integers.
  - Input an operation (+, -, *, /).
  - Perform the operation on two integers and output result.

Practice

- Practice 1.4:

  Write algorithm to rank a student as follow:

  - Input literature, math, physics points.
  - Calculate student GPA.
  - Rank student based on GPA:
    + Excellent: GPA >= 8.5.
    + Good: GPA >= 7.0.
    + Fair: GPA >= 5.0.
    + Failed: GPA < 5.0.
  - Output GPA and rank.

Practice

- Practice 1.5:

  Write algorithm to count number of days in a month as follow:

  - Input a month and a year.
  - Count number of days in that month and output result.

Practice

- Practice 1.6:

  Write algorithm to compute exponentiation of an integer as follow:

  - Input x and n.
  - Compute x^n and output result.

Practice

- Practice 1.7:

  Write algorithm to do the following:

  - Input 2 integers x, n (n > 0).
  - Calculate and output $S = x + x^2 + \ldots + x^n$.

Practice

- Practice 1.8:
  - Set up a C/C++ programming environment on your computer.

Practice

- Practice 1.9:

  Edit, compile and run the following C/C++ program.

  What does the program ask for?

  What does the program print to screen?

```
#include <stdio.h>

int main()
{
	int   a, b, c, max;
	printf("Enter 3 integers = ");
	scanf("%d %d %d", &a, &b, &c);
	max = a;
	if (b > max) max = b;
	if (c > max) max = c;
	printf("max = %d", max);
}
```

Practice

- Practice 1.10:

Edit, compile, run and debug the following C/C++ program.

a) What does the program ask for and print to screen?

b) When enter n = 10, list the values of s when n = 3.

```
#include <stdio.h>

int main()
{
	int		n;
	long long	s;
	printf("Enter an integer = ");
	scanf("%d", &n);
	for (s = 1; n > 0; n--)
		s = s * n;
	printf("Factorial = %lld ", s);
}
```

Q
&
A